MAIS PAC

IPhO 02.15: Thermodynamics and the Ideal Gas · Worked Solutions

# Thermodynamics and the Ideal Gas: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IPhO questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

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**Solution 1 — The ideal gas law.** Return →

(a) T = 300 {K}, so V = (nRT)/(p) = ((0.25)(8.31)(300))/(1.5×10^5) = 4.2×10^(-3) {m³} = 4.2 {L}.

(b) N = nN_A = (0.25)(6.02×10^(23)) = 1.5×10^(23) molecules.

(c) At constant T, p∝1/V, so halving the volume doubles the pressure: 3.0×10^5 {Pa}.
**Check:** pV = nRT = 623 {J} at both states, as Boyle's law requires.

**Solution 2 — Molecular speeds.** Return →

(a) v_({rms}) = √(3RT/M) = √(3(8.31)(300)/(4.0×10^(-3))) = √(1.87×10^6) = 1.37×10³ {m,s^(-1)}.

(b) v_({rms}) = √(3(8.31)(300)/(28×10^(-3))) = √(2.67×10^5) = 5.17×10² {m,s^(-1)}.

(c) v_({rms})∝√(T/M), so equal speeds need T_({N_2}) = 300×(28/4.0) = 2.1×10³ {K}.
**Check:** helium is seven times lighter, so at the same temperature its molecules move √(7) = 2.65 times faster; 1370/517 = 2.65.

**Solution 3 — Work in three processes.** Return →

(a) Isothermal: W = nRTln (V_2/V_1) = (1)(8.31)(300)ln 2 = 1.73×10³ {J}.

(b) Isobaric from the same state: p = nRT/V_1 = (1)(8.31)(300)/(0.010) = 2.49×10^5 {Pa}, so W = pΔ V = (2.49×10^5)(0.010) = 2.49×10³ {J}, and T_2 = 2T_1 = 600 {K}.

(c) In the isothermal case the pressure falls as the gas expands, so the average pressure is lower and less work is done. The isobaric process needs more heat, because it must both do more work and raise the temperature (Δ U = (3)/(2)nRΔ T = 3.7 {kJ}). **Check:** isothermal Q = W = 1.73 {kJ}; isobaric Q = (5)/(2)nRΔ T = 6.2 {kJ}, consistent with more work plus more internal energy.

**Solution 4 — An adiabatic compression.** Return →

(a) pV^γ is constant with γ = 5/3, so

$$p_2 = p_1((V_1)/(V_2))^{(5/3)} = (1.0\times10^5)(8)^{(5/3)} = (1.0\times10^5)(32) = 3.2\times10^6 \text{ {Pa}}. \quad (1)$$

(b) TV^(γ-1) is constant: T_2 = 300×8^(2/3) = 300(4) = 1200 {K}.

(c) W_({on}) = Δ U = (3)/(2)nRΔ T = (3)/(2)(2.0)(8.31)(900) = 2.2×10^4 {J}. **Check:** the pressure rose by a factor 32 = 8^(5/3) and the temperature by 4 = 8^(2/3), and pV/T is the same at both ends: (10^5)(V)/(300) = (3.2×10^6)(V/8)/(1200).

**Solution 5 — A Carnot engine.** Return →

(a) _C = 1 - 300/500 = 0.40, so W_(max) = 0.40×2.0 = 0.80 {kJ}.

(b) Q_C = Q_H - W = 2.0 - 0.80 = 1.2 {kJ}.

(c) With = 0.30, W = 0.60 {kJ}, so Q_C = 1.4 {kJ}. The entropy changes are Δ S_H = -2000/500 = -4.0 {J,K^(-1)} (reservoir) and Δ S_C = +1400/300 = +4.67 {J,K^(-1)}, so the universe gains 0.67 {J,K^(-1)} per cycle. **Check:** for the reversible (Carnot) engine the two changes are equal and opposite, 0.67 {J,K^(-1)} less, exactly as the second law requires.

**Solution 6 — Kinetic theory estimates.** Return →

(a) n_V = (p)/(k_BT) = (1.0×10^5)/((1.38×10^(-23))(300)) = 2.4×10^(25) {m^(-3)}.

(b) ≈ (1)/(√(2),n_Vπ d²) = (1)/(1.414(2.4×10^(25))π(3.0×10^(-10))²) = 1.0×10^(-7) {m} — about 100 {nm}.

(c) v_({rms}) = √(3k_BT/m) = √(3(1.38×10^(-23))(300)/(4.8×10^(-26))) = 5.1×10² {m,s^(-1)}, so the collision frequency is v/ = 5.1×10^9 {s^(-1)} — five billion collisions per second. **Check:** this is why diffusion in air is slow compared with the molecular speed: each molecule zigzags a huge distance to make progress of a centimetre.

**Solution 7 — Mixing and entropy.** Return →

(a) No work is done and no heat crosses the insulation, so U is unchanged and T stays the same. The volume doubles and the amount of gas doubles, so p is unchanged as well: the gas simply fills twice the space at the same density.

(b) Considering the **whole** gas, it expands from V to 3V? No: the two halves already occupy V and 2V; after mixing each molecule has access to 3V. For each mole,

$$\Delta S = nR\ln ()/(){V_f}{V_i} = 4.6\ln 3 \ \{\text{for the first}\} + 4.6\ln ()/(){3}{2} \ \{\text{for the second}\} = 5.0 + 1.9 = 6.9 \ \{J,K^{(-1)}\}. \tag{2}$$

(c) The reverse — all the gas spontaneously gathering in one half — is not forbidden by energy conservation, but the number of microscopic arrangements corresponding to the mixed state is enormously larger. The second law is statistical, and the probability of the reverse is of order 2^(-N) with N10^(23). **Check:** Δ S > 0 and Δ U = 0: entropy and energy are independent bookkeeping, which is precisely the content of the second law.

**Solution 8 — A cyclic process.** Return →

(a) Work is done only in the two isothermal steps:

$$W_1 = nR(400)\ln 2 = 2.31 \ \{kJ\}, \ W_3 = nR(300)\ln ()/()12 = -1.73 \ \{kJ\}, \tag{3}$$

so the net work is 0.58 {kJ}.

(b) Step 1: Q_1 = W_1 = +2.31 {kJ}. Step 2 (isochoric cooling from 400 to 300 K): Q_2 = (3)/(2)nR(-100) = -1.25 {kJ}. Step 3: Q_3 = W_3 = -1.73 {kJ}. Step 4 (isochoric heating): Q_4 = +1.25 {kJ}. The net heat is 2.31 - 1.25 - 1.73 + 1.25 = 0.58 {kJ}, matching the net work.

(c) = W/Q_({in}) = 0.58/2.31 = 0.25. The Carnot efficiency between 400 and 300 K is 1 - 300/400 = 0.25 — equal, because this cycle is reversible. **Check:** the two isochoric steps exchange equal and opposite heat, so they do not affect the efficiency, only the peak temperature.

**Solution 9 — The speed of sound.** Return →

(a) For an adiabatic compression, dp/dρ = γ p/ρ, and the wave equation v = √(dp/dρ) gives v = √(γ p/ρ). Using ρ = pM/(RT) gives v = √(γ RT/M).

(b) v = √(1.40(8.31)(293)/0.029) = √(1.18×10^5) = 343 {m,s^(-1)}.

(c) At fixed T, ρ∝ p, so p/ρ is constant and the pressure cancels. Humidity matters because water vapour has molar mass 18 against 29 for air, so moist air is **less** dense: the speed of sound rises slightly on a humid day. **Check:** 343 {m,s^(-1)} is the standard value, and the ratio to the rms molecular speed is about 0.7, as expected for a wave carried by collisions.

**Solution 10 — A real engine.** Return →

(a) Using the hottest and coldest temperatures, _C = 1 - 400/2200 = 0.82.

(b) = W/Q_H = 60/200 = 0.30.

(c) The Carnot value assumes the whole heat input arrives at the top temperature and the whole output leaves at the bottom one, with no friction and infinitely slow processes. In a real engine the combustion gases cool as they expand, the exhaust leaves far above ambient, there is friction and turbulence, and the cycle takes a finite time. The gap between 30% and 82% is the price of speed, materials and irreversibility. **Check:** even a very good power station reaches only about 45% of its Carnot limit, so 30% against 82% is typical of a car engine rather than a sign of a mistake.